Exploratory Data Analysis (EDA) & Feature Engineering Report

1. Introduction & Data Engineering Pipeline As part of the initial phase of this project, the objective was to construct a preprocessing pipeline that extracts structured numerical features from detected image regions. Rather than utilizing end-to-end deep learning pipelines, this approach converts raw aerial LWIR drone images from the AMLID dataset into a tabular dataset suitable for classical statistical modeling.

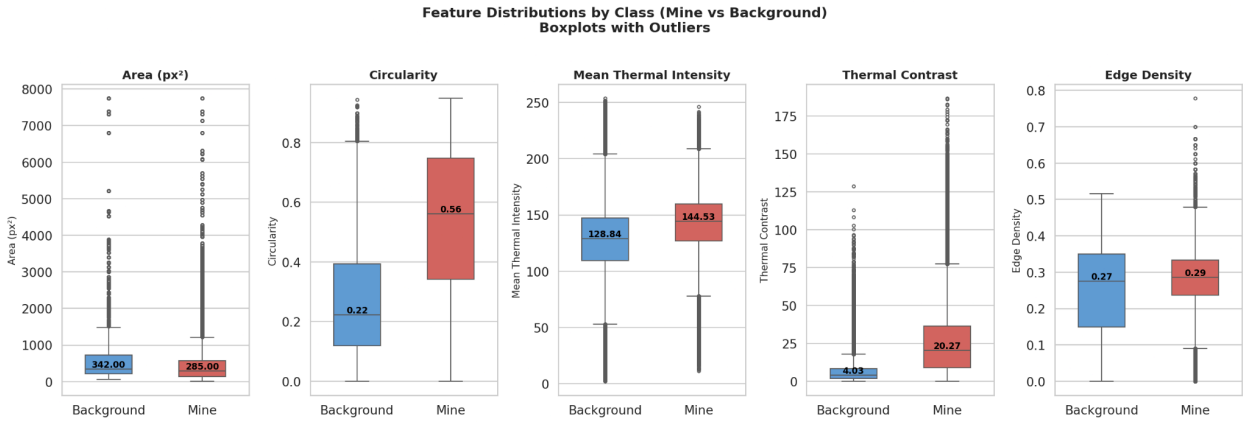
A robust feature extraction script was developed to process the bounding box annotations, isolating candidate objects and extracting five handcrafted features: object area, circularity / compactness, mean thermal intensity, thermal contrast with background, and edge density.

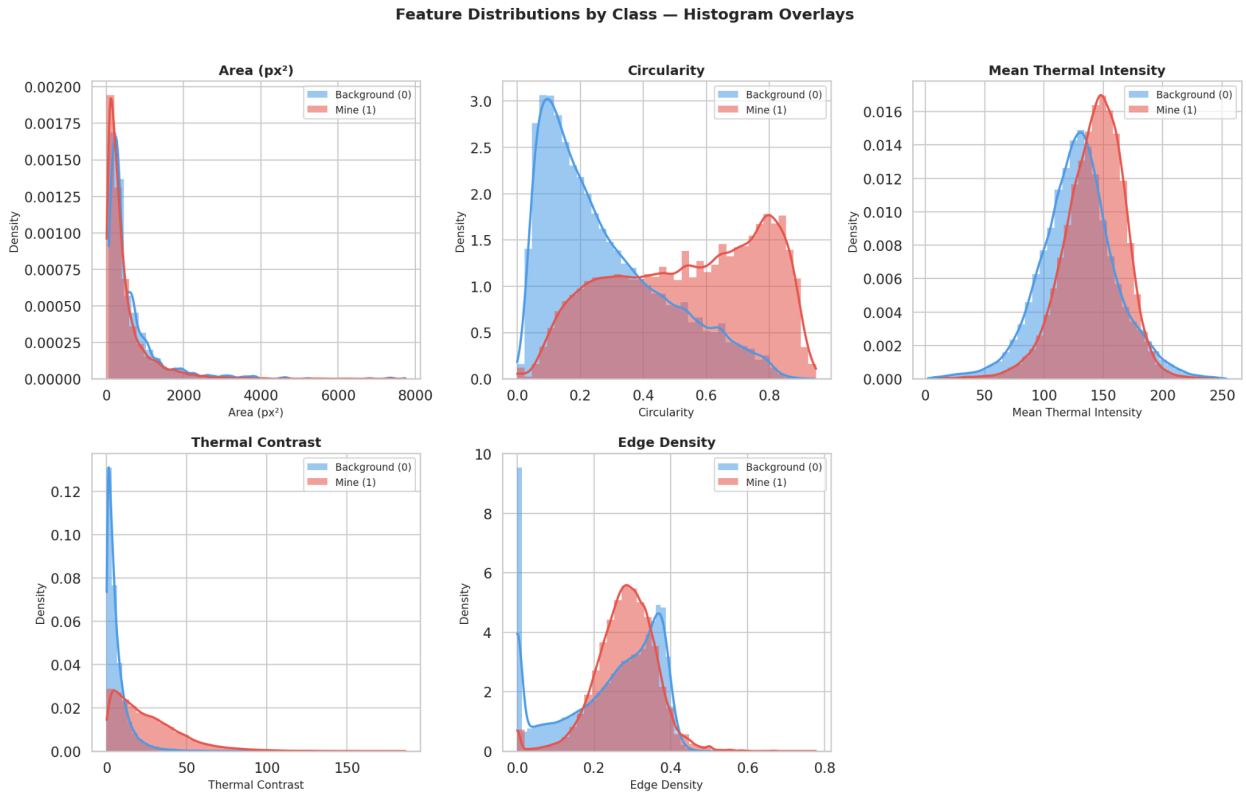
To allow deeper statistical investigation and robust evaluation, the problem is restricted to binary classification (mine vs. non-mine). Background patches (Label = 0) were sampled from the same images to closely match the number of annotated mines (Label = 1), resulting in a virtually perfectly balanced dataset. The final dataset contains 229,384 total samples (115,093 mines and 114,291 background patches), ensuring robust statistical validity with zero missing values.

2. Exploratory Data Analysis (EDA)

A. Feature Distributions and Outlier Detection

To understand the predictive power and behavior of the engineered features, boxplots and Kernel Density Estimation (KDE) overlaid histograms were generated.



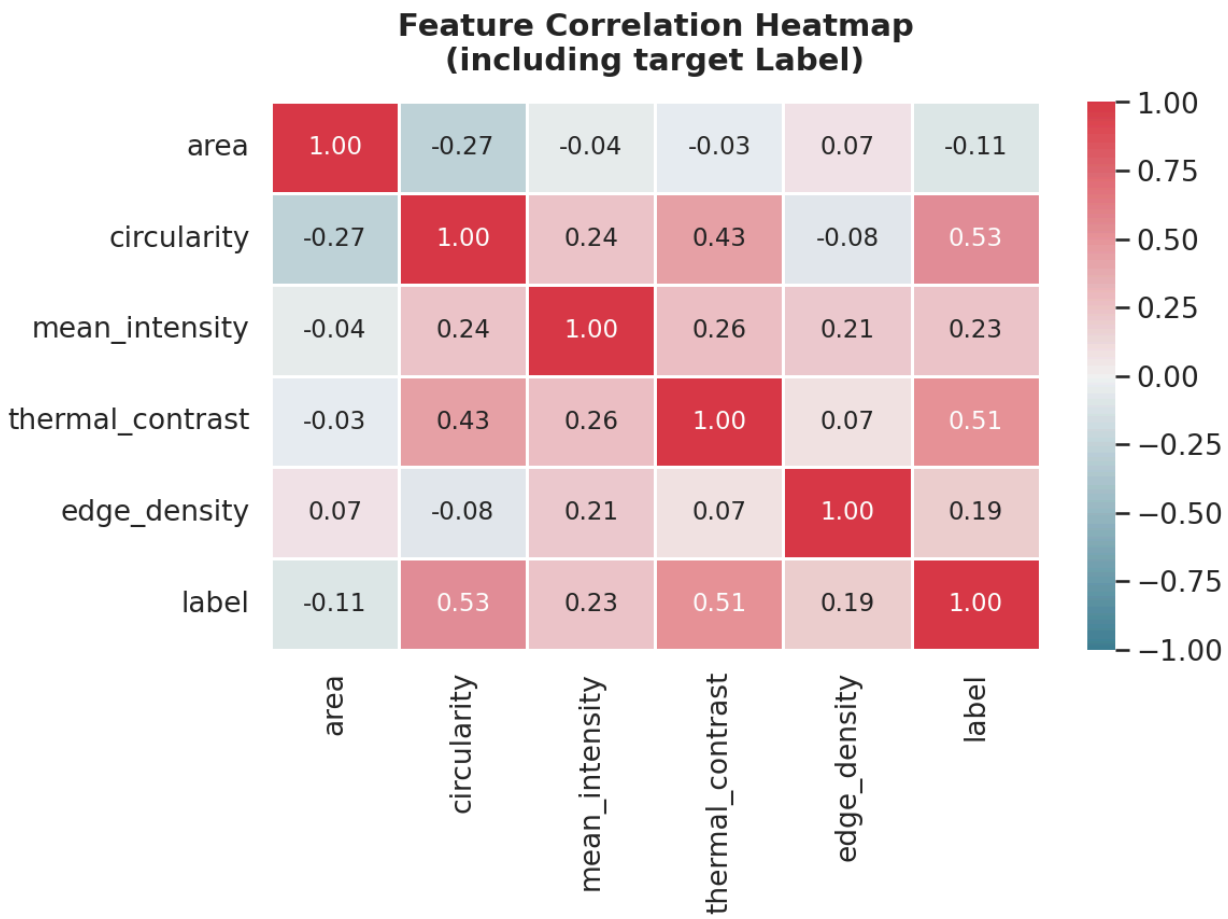


Key Insights from Distributions:

- **Circularity / Compactness:** The visualizations demonstrate that circularity provides a highly effective decision boundary between the classes. Mines exhibit a significantly higher mean circularity compared to the irregular contours of background patches. This aligns with the physical reality of manufactured anti-tank and anti-personnel mines.
- **Thermal Contrast:** There is a distinct bimodal separation in thermal contrast. Mines show substantially higher thermal contrast with their background compared to random soil or vegetation patches, confirming that landmines absorb and emit thermal radiation differently in LWIR imagery.

B. Correlation Analysis & Multicollinearity

A Pearson correlation matrix was computed to evaluate the linear relationships between the handcrafted features and the target label, as well as to check for multicollinearity among the predictors.



Key Insights from Correlation:

- **Strong Predictors:** Circularity (r = +0.525) and Thermal Contrast (r = +0.509) hold the strongest positive correlations with the target variable, validating the feature engineering strategy. Area proved to be a weak, slightly inverse predictor (r = -0.108).
- **Feature Independence:** Crucially, the inter-feature correlation remains low across the board. This lack of strong multicollinearity is optimal for classical modeling, ensuring that each feature contributes unique variance to the model without inflating standard errors.

3. Conclusion and Next Steps The data engineering phase successfully yielded a clean, balanced, and highly discriminative tabular dataset of over 229,000 samples. The extracted features demonstrate strong predictive potential. The dataset is now fully prepared for the modeling phase, where baseline implementation and regularization experiments (L_1 vs. L_2) will be conducted to evaluate predictive performance.